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## Submitter Information

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## General Comment

Applied Intuition welcomes the opportunity to respond to the National Science Foundation and the Networking and Information Technology Research and Development National Coordination Office's Request for Information on "Development of an Artificial Intelligence Action Plan." Please see the attached comment.

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## Attachments

Applied Comments on NSF_FRDOC_0001-3479 AI Action Plan

March 14, 2025

Dr. Lynne Parker
Principal Deputy Director
Office of Science and Technology Policy
Executive Office of the President
Eisenhower Executive Office Building
1650 Pennsylvania Avenue
Washington, D.C. 20504

**Re: Docket No. NSF_FRDOC_0001-3479 "Development of an Artificial Intelligence Action Plan"**

Dear Dr. Parker,

Applied Intuition welcomes the opportunity to respond to the National Science Foundation (NSF) and the Networking and Information Technology Research and Development National Coordination Office's (NITRD NCO) Request for Information (RFI) on "Development of an Artificial Intelligence Action Plan." As a company focused on improving safety through advanced artificial intelligence (AI), Applied Intuition appreciates the ability to provide context for how AI-defined technologies, including autonomous vehicles (AVs) and advanced driver-assistance systems (ADAS), can revolutionize our defense and transportation systems.

Applied Intuition is a vehicle software supplier that accelerates the adoption of safe and intelligent machines. Founded in 2017, the company delivers the AI-powered ADAS/AV toolchain, vehicle platform, and autonomy software to eighteen of the top twenty, non Chinese, global automakers. Applied Intuition serves the automotive, defense, trucking, construction, mining, and agriculture industries. The company is headquartered in Mountain View, CA, with offices in Ann Arbor and Detroit, MI, Destin, FL, San Diego, CA, Washington, DC, Stuttgart, Munich, Stockholm, Seoul, and Tokyo.

Applied Intuition Defense, a business unit of Applied Intuition, is a strategic software supplier that accelerates the deployment of next-generation autonomous capabilities to the warfighter. Today, Applied Intuition Defense provides rapid, agile software development best practices from the commercial automotive industry to major programs across the Department of Defense (DOD). Current defense customers include the Air Force, Navy, Army, DARPA, and the Chief Digital and AI Office (CDAO). Applied recently acquired EpiSci, a leader in AI and trusted autonomy software, which positions Applied Intuition Defense as the premier autonomy software developer across all domains-land, air, sea, and space.

In the RFI, the NITRD NCO seeks further information on the highest priority policy actions that should be in the new AI Action Plan. As a dual-use AI company, Applied Intuition supports the Administration's efforts to "enhance America's position as an AI powerhouse and prevent unnecessarily burdensome requirements from hindering private sector innovation."[1] Highlighted below are critical applications of AI technologies that should be prioritized and corresponding policy recommendations.

## I. Opportunities of AI in Defense

AI has critical national security benefits for the U.S. military such as supporting rapid analysis in intelligence; providing early threat warnings through advanced sensing; utilizing autonomous vehicles for military logistics; and enabling human-machine teaming. Autonomous systems, which rely on AI and machine learning (ML) technologies, can increase mission effectiveness, reduce collateral damage, lower costs, offset personnel shortages, and increase warfighter safety. Autonomous capabilities that are properly trained and tested have unique advantages since computers can operate without fear, bias, or fatigue, reducing the likelihood of fatal decisions.

In the future, any high-end fight with a near-peer adversary will involve a rapidly changing battlespace, a vast number of targets, and new, never-before-seen autonomous platforms with software deployed to the edge.[2] To maximize these national security benefits, AI-empowered autonomous systems must be continuously tested and validated to ensure their safety, trustworthiness, and performance before they are fielded. Thousands of "dumb drones" are easily defeated; fully-networked, autonomous, and rapidly upgradable drones ensure our warfighters will be able to continually adapt to evolving battlefield conditions. A key example of the impact of AI software is the war in Ukraine. Russian forces placed tires on their aircraft in an attempt to spoof loitering munitions.[3] Without a rapid AI retraining pipeline, Ukraine's perception software would have been unable to recognize and destroy Russian assets.

Virtual testing software enables users to evaluate the performance of an autonomy stack in limitless scenarios that would otherwise be too difficult, unsafe, or costly to replicate in real-world testing. The AI Action Plan should support the DOD's utilization of an enterprise-level, all-domain software development and testing pipeline to drive trusted AI and autonomy adoption at the speed of relevance.

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[1] *Public Comment Invited on Artificial Intelligence Action Plan*, The White House, (Feb. 25, 2025), https://www.whitehouse.gov/briefings-statements/2025/02/public-comment-invited-on-artificial-intelligence-action-plan/.
[2] U.S. Department of Defense, *2022 National Defense Strategy, Nuclear Posture Review, and Missile Defense Review*, 4.
[3] Helen Regan and Irene Nasser, "Russia Is Putting Car Tires on Aircraft to Protect Them from Ukrainian Drone Attacks," *CNN*, September 6, 2023, https://www.cnn.com/2023/09/06/europe/russia-aircraft-car-tires-ukraine-drones-intl-hnk/index.html.

## II. Opportunities of ADAS and AVs in Transportation

ADAS and AV technologies represent an important application of AI-enabled technologies that are poised to revolutionize transportation.

### a. *ADAS*

ADAS technologies assist drivers in a number of ways, though they do not take full control of the vehicle and always require human monitoring. These AI-powered technologies include blind spot and forward collision warnings, lane departure warnings, rear collision warnings, automatic emergency braking (AEB) and pedestrian AEB (PAEB), adaptive cruise control, and lane centering and lane keeping assistance.[4]

Utilizing advanced AI algorithms, ADAS technologies can improve roadway safety by reducing the possibility of human error and providing a driver with warnings and course corrections they may otherwise have missed or failed to take. A NHTSA study showed that lane keeping assist-equipped vehicles are 24 percent less likely to be involved in fatal road departure crashes compared to non-equipped models.[5] The wider adoption of ADAS can prevent over 20,000 deaths a year, and prevent or mitigate 1.69 million injuries, based on an analysis by the National Safety Council.[6]

ADAS technologies are growing increasingly common in passenger and commercial vehicles. The National Highway Traffic Safety Administration (NHTSA) finalized a rule that mandates the inclusion of AEB and PAEB in new light-duty vehicles starting in 2029,[7] and has proposed a similar mandate for heavy-duty vehicles.[8] These mandates will greatly expand the deployment of ADAS technologies and bring the safety benefits of the technology to millions of Americans.

### b. *AVs*

AVs are vehicles equipped with an automated driving system (ADS) capable of performing the entire dynamic driving task on a sustained basis.[9] ADS-equipped vehicles

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[4] *Driver Assistance Technologies*, National Highway Traffic Safety Administration, https://www.nhtsa.gov/vehicle-safety/driver-assistance-technologies (last visited March 14, 2025).

[5] H. Scharber, *Estimating Effectiveness of Lane Keeping Assist Systems in Fatal Road Departure Crashes*, DOT HS 813 663 (Dec. 2024) https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813663.

[6] National Safety Council, *Occupant Protection: Advanced Driver Assistance System*, NSC Injury Facts, https://injuryfacts.nsc.org/motor-vehicle/occupant-protection/advanced-driver-assistance-systems/data-details/ (last visited May 16, 2024).

[7] Federal Motor Vehicle Safety Standards; *Automatic Emergency Braking Systems for Light Vehicles*, 89 Fed. Reg. 39686 (May 9, 2024) (to be codified at 49 C.F.R. pts. 571, 595, 596).

[8] *Heavy Vehicle Automatic Emergency Braking; AEB Test Devices*, 88 Fed. Reg. 43174 (Sept. 5, 2023).

[9] *SAE International, Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles*, J2016_202104 (2021), https://www.sae.org/standards/content/j3016_202104/.

navigate the world with a suite of sensor systems, including cameras, radar, and lidar, using AI to process the collected data to help plan the vehicle's route, avoid obstacles, and safely interact with other vehicles and pedestrians.

With nearly 40,000 people killed in motor vehicle traffic incidents last year[10] and pedestrian deaths on the rise,[11] significant improvements to roadway safety are needed now more than ever. By removing the potential for human error, whether due to fatigue, intoxication, or distraction, AVs are positioned to reduce roadway deaths and injuries. NHTSA's own research has shown that 55% of all people injured or killed in fatal accidents tested positive for one or more drugs, including alcohol.[12] Distraction from cell phones is another major factor in roadway incidents; drivers using handheld cell phones are two to six times more at risk for a crash.[13] AVs, which are incapable of driving drunk, distracted, or otherwise impaired, can reduce roadway deaths and other incidents.

When properly tested using modeling and simulation, in addition to real-world testing, ADAS and AVs are vital applications of AI-enabled transportation technologies and are worthy of inclusion in the AI Action Plan.

## III. Policy Recommendations

Advanced AI technologies like AVs need additional regulatory certainty to improve the competitiveness of the domestic industry. Current regulatory environments are often very restrictive in an effort to address each system's risks individually. The following recommendations would provide clear guidance to U.S. industry, increase roadway safety, ande also effectively deregulate certain burdensome aspects of the current regulatory environment that are stifling innovation.

### a. *DOD Manhattan Project for Autonomy*

Despite the fact that software-defined autonomous systems are redefining the modern battlefield, the U.S. DOD is underinvesting in autonomy. Despite pressure on service programs to respond to this dramatic shift, the DOD struggles to deliver production-ready autonomous platforms to U.S. warfighters. Meanwhile, DOD's existing AI investments are overindexed for workflow efficiencies, such as the use of large language models (LLMs), instead of enabling

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[10] National Highway Traffic Safety Administration, *Early Estimate of Motor Vehicle Traffic Fatalities for the First 9 Months (January–September) of 2024*, https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813670.
[11] Governors Highway Safety Associations, *Pedestrian Traffic Fatalities by State January - June 2023 Preliminary Data, (2023)*, https://www.ghsa.org/resources/Pedestrians24.
[12] National Highway Traffic Safety Administration, U.S. Department of Transportation, DOT HS 813 399, Alcohol and Drug Prevalence Among Seriously or Fatally Injured Road Users, (2022), https://rosap.ntl.bts.gov/view/dot/65623/dot_65623_DS1.pdf.
[13] *Distracted driving*, Insurance Institute for Highway Safety, https://www.iihs.org/topics/distracted-driving (last visited March 14, 2025).

autonomous platforms with tactical capabilities. Furthermore, the data to enable command and control workflow improvements will not exist without software-defined and autonomous capabilities collecting data at the tactical edge.

Therefore, the AI Action Plan should support efforts to establish a "Manhattan Project for Defense Autonomy" with robust funding and authority to deploy software systems at Silicon Valley speed. This effort should harness a software development and testing pipeline to ensure interoperability of autonomous systems across domains and vendors, while also ensuring software can be rapidly updated and tested. Not only will this require significant funding, it will require special hiring authorities to staff up quickly including those from industry.

b. *Software acquisition modernization*

Given the importance of software on the modern battlefield, it is critical that DOD not treat software the same as hardware. Software is never finished. It continues to live and grow as conditions rapidly change. Unlike hardware, its evolutions occur in weeks, not years, and if the software does not evolve at this speed of relevance, it becomes obsolete. DOD recognizes this fact in its Software Modernization Strategy, which states: "fighting and winning on the next battlefield will depend on DOD's proficiency to rapidly and securely deliver resilient software capabilities."[14] Applied also supports the intent of Secretary Hegseth's memo, "Directing Modern Software Acquisition to Maximize Lethality," issued March 6, 2025 to accelerate software acquisition across the enterprise .[15]

Currently, many of DOD's largest legacy investments are dependent on a single vendor. As a result of past issues with "vendor-lock," DOD often seeks to own the software it procures. This overreaction is counterproductive because the agency lacks the incentive to continuously improve and innovate the software it owns and thus relies on outdated software. Instead of owning the software, Applied recommends that DOD owns the data it collects from its sensors. This data will enable DOD to develop a common operating picture that can be shared across domains and areas of responsibility. Conversely, DOD should license software from multiple commercial vendors to ensure that DOD gets the best software based on evolving battlefield conditions.

[14] U.S. Department of Defense. *Department of Defense Software Modernization Strategy*. February 3, 2022. https://media.defense.gov/2022/Feb/03/2002932833/-1/-1/1/DEPARTMENT-OF-DEFENSE-SOFTWARE-MODERNIZATION-STRATEGY.PDF.

[15] U.S. Department of Defense, *Modern Software Acquisition to Speed Delivery, Boost Warfighter Lethality*, (March. 10, 2025), https://www.defense.gov/News/News-Stories/Article/Article/4114775/modern-software-acquisition-to-speed-delivery-boost-warfighter-lethality/.

The AI Action Plan should support modernizing the means by which DOD acquires software, such as splitting defense hardware & software acquisition for AI-enabled unmanned systems, and owning its data while licensing commercial software.

c. *National AV Framework*

AV companies are currently burdened by a patchwork of varying state laws for testing and deployment on U.S. roads. Applied strongly supports establishing a national framework that sets standards for the design, construction, and performance of AVs and preempts state laws. This national framework should also enhance the U.S. DOT's current data collection efforts beyond the current Standing General Order to preempt varying state requirements and to improve safety across the industry. The framework should also encourage NHTSA to modernize its test infrastructure to ensure AV and ADAS systems are deployed safely.

## IV. Conclusion

Applied Intuition appreciates the chance to provide context on the opportunities for AI-enabled technologies in defense and transportation. We look forward to continued engagement with the Administration to provide subject-matter expertise on the benefits and opportunities of AI technologies. This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Sincerely,

Nicholas Kazvini-Gore
Head of Government Affairs, Applied Intuition